

ST2MML - 01

Machine Learning

Welcome
February 7, 2025

Discord



- **#efrei-st2mml**
- <https://discord.gg/JXQcPy3amM>



Who am I?



Alexis Perrier

- PhD TelecomParis 95'
- Data scientist, author, teacher
- Sci-fi, biking, LLMs,
- alexis.perrier@gmail.com



Started working with mysql in 2000

Teaching : SciencePo, IPSA, Eifrei, Epita, Gustave Eiffel, Openclassrooms, ...

The Course

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG
PILE OF LINEAR ALGEBRA, THEN COLLECT
THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL
THEY START LOOKING RIGHT.



Intro to Machine Learning - Syllabus

Models

- Linear Regression
- Logistic Regression
- Stochastic gradient descent
- The perceptron
- Feed-forward Neural Networks
- K-means Clustering
- Decision trees
- Random Forests
- XGBoost

Methods

- supervised : Regression, Classification
- non-supervised: Clustering
- Model regularization
- Model performance evaluation
- Metrics
- Bias and overfitting,
- Bagging & Boosting
- Imbalanced Datasets

Concepts

- Cost Function
- bias-variance decomposition
- Back Propagation
- Activation functions

Intro to Machine Learning - Tools

- python
 - pandas
 - scikit-learn
 - keras
- notebooks, Colab
- scripts
- github
- using the terminal

Schedule

CODE	LIBELLÉ	MODE	DATE	J.	HEURE DÉBUT	HEURE FIN	DURÉE	MODALITÉ
ST2MML	Machine Learning	CTP	07/02/2025	ven	09:10	13:40	04:00	PRÉS.
ST2MML	Machine Learning	CTP	14/02/2025	ven	09:10	12:30	03:00	PRÉS.
ST2MML	Machine Learning	CTP	21/02/2025	ven	09:10	12:30	03:00	PRÉS.
ST2MML	Machine Learning	CTP	07/03/2025	ven	09:10	12:30	03:00	PRÉS.
ST2MML	Machine Learning	CTP	11/03/2025	mar	08:00	12:30	04:00	PRÉS.
ST2MML	Machine Learning	CTP	14/03/2025	ven	09:10	12:30	03:00	PRÉS.
ST2MML	Machine Learning	CTP	17/03/2025	lun	13:50	18:20	04:00	PRÉS.
ST2MML	Machine Learning	CTP	18/03/2025	mar	13:50	18:20	04:00	PRÉS.

Course Philosophy

- Hands on
- don't waste your time
- no pain no gain
- LLMs are our friends

Tools

- [Google Drive](#): slides & colab notebooks
- Github: course material <https://github.com/SkatAI/efrei-st2mml>
- Discord: questions and course material <https://discord.gg/JXQcPy3amM>

All course material is available in the discord channel and on github

Prerequisites

- Know your laptop (Path, install, etc)
- Terminal and command line
 - Mac : iTerm
 - Windows :
 - git console
 - powershell Admin

AI policy

LLMs are here to stay, we need to learn how to work with LLMs

- How to avoid
 - **cognitive offloading**
 - rabbit hole
 - hallucinations
- Available LLMs
 - [DeepSeek](#)
 - Gemini on [aistudio.google.com](#) and in [Colab](#)
 - Claude.ai
 - OpenAI's ChatGPT.com
 - [Grok](#)
 - [Qwen.ai](#)

AI policy

**NO PAIN
NO GAIN**

You don't become fit watching reels on Insta

Don't let the LLM do the work for you

Let's avoid

Teacher has the LLM create the work

Student has the LLM do the work

Teacher has the LLM grade the student's work

Course Evaluation

- Lab works, projects => 50%
- Final exam / partiel => 50%

Getting to know you

Please fill out this form

We'd like to know a bit more about you

So that we can adapt the course to meet your expectations

All questions are optional (except your email)

<https://forms.gle/1ksP3qtAk2F5N2gT8>

Google Colab

Google Colab

Like google docs but also for executing code
Shareable, collaborative work

A notebook is a series of **executable** **cells**

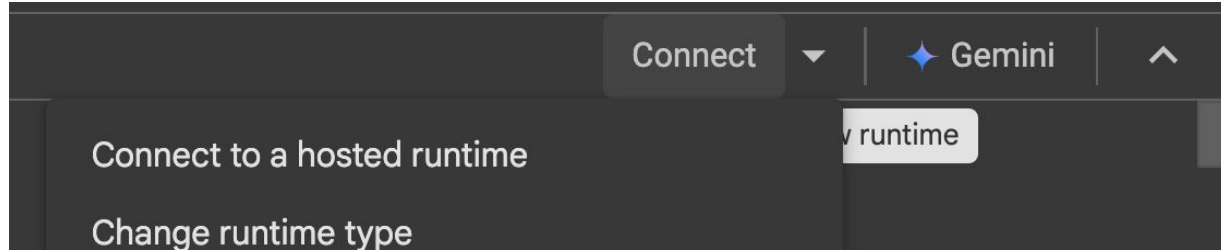
- code (python)
- text with Markdown



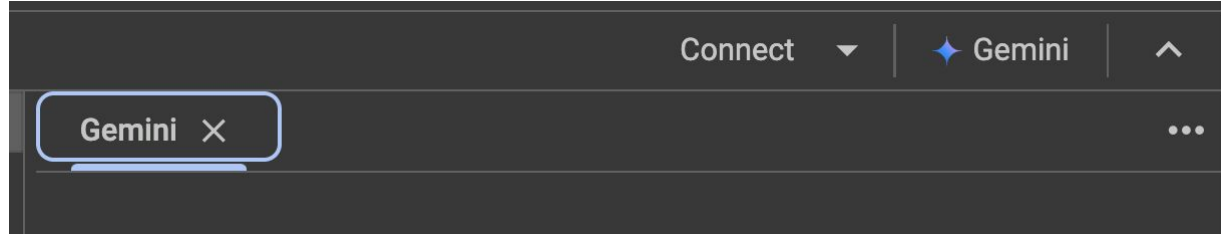
<https://colab.research.google.com/>

Google Colab

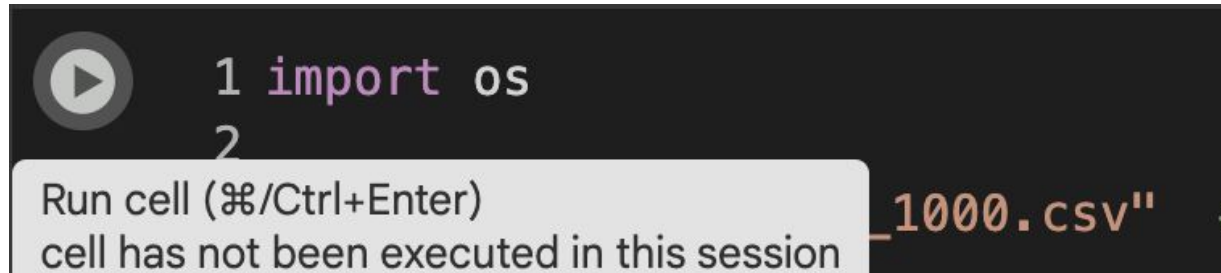
Connect to a runtime



Prompt Gemini to write the code for a given task



Click to run the code



Text with Markdown

	
<pre># LLM ## Definition A large language model (LLM) is a type of machine learning model designed for [natural language processing](https://en.wikipedia.org/wiki/Natural_language_processing) tasks such as language generation. ## History of NLP - 1954s: The Georgetown experiment - some years - 2010, Tomáš Mikolov applied a simple recurrent neural network with a single hidden layer to language modelling. ## A quote: > It is not the answer that enlightens, but the question > Ionesco </pre>	LLM Definition A large language model (LLM) is a type of machine learning model designed for natural language processing tasks such as language generation. History of NLP • 1954s: The Georgetown experiment • some years • 2010, Tomáš Mikolov applied a simple recurrent neural network with a single hidden layer to language modelling. A quote: It is not the answer that enlightens, but the question Ionesco

Text with Markdown

What you write

Simple Syntax

Header 1

****this is bold****, not bold

[a link](https://efrei.fr)

What you get !

Simple Syntax

Header 1

this is bold, not bold

[a link](https://efrei.fr)